EXERCISE 1

Constant-Coefficient Equations with Forcing (Eleven IVPs)

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Undetermined Coefficients • 11 IVPs • Resonance

STATEMENT

Find the general solution of the following differential equations. You will find it useful to look up Exercise 4 of Chapter 3, part 1. For each item (n), find the unique solution such that $x(0) = n$ and $\dot{x}(0) = 0$. E.g. for item (iii), $x(0) = 3$, etc.

- (i) $\ddot{x} - 4x = t^2$;
- (ii) $\ddot{x} - 4\dot{x} = t^2$;
- (iii) $\ddot{x} + \dot{x} - 2x = 3e^{-t}$;
- (iv) $\ddot{x} + \dot{x} - 2x = e^{t}$;
- (v) $\ddot{x} + 2\dot{x} + x = e^{-t}$;
- (vi) $\ddot{x} + \omega^2 x = \sin(\alpha t)$ for $\alpha \neq \omega$;
- (vii) $\ddot{x} + \omega^2 x = \sin(\alpha t)$ for $\alpha = \omega$;
- (viii) $\ddot{x} + 2\dot{x} + 10x = e^{-t}$;
- (ix) $\ddot{x} + 2\dot{x} + 10x = e^{-t}\cos(3t)$;
- (x) $\ddot{x} + 6\dot{x} + 10x = e^{-3t}\cos t$,
- (xi) $\ddot{x} + 4\dot{x} + 4x = e^{2t}$;

PHASE 1: SETUP, CLASSIFICATION AND DATA

All eleven equations are **linear, second-order, inhomogeneous equations with constant coefficients**, $L[x] = \ddot{x} + b\dot{x} + cx = f(t)$ (the coefficient of $\ddot{x}$ is 1). The forcing f is built from polynomials, exponentials and sinusoids, so the method of undetermined coefficients applies (Robinson Ch. 14). Hypotheses: for (vi) and (vii) the parameters $\omega > 0$ and α are real, with $\alpha \neq \omega$ in (vi) and $\alpha = \omega$ in (vii).

The coefficients are continuous on all of $\mathbb{R}$, so for every pair of initial data there is exactly one solution, defined for all t (existence and uniqueness, Robinson Thm 11.1 and Thm 16.1). The data of item (n) are $x(0) = n, \dot{x}(0) = 0$, with n the position of the item in the list:

Item	Equation $L[x]=f$	$x(0)$	$\dot{x}(0)$
(i)	$\ddot{x}-4x=t^2$	1	0
(ii)	$\ddot{x}-4\dot{x}=t^2$	2	0
(iii)	$\ddot{x}+\dot{x}-2x=3e^{-t}$	3	0
(iv)	$\ddot{x}+\dot{x}-2x=e^{t}$	4	0
(v)	$\ddot{x}+2\dot{x}+x=e^{-t}$	5	0
(vi)	$\ddot{x}+\omega^2 x=\sin(\alpha t)$ for $\alpha\neq\omega$	6	0
(vii)	$\ddot{x}+\omega^2 x=\sin(\alpha t)$ for $\alpha=\omega$	7	0
(viii)	$\ddot{x}+2\dot{x}+10x=e^{-t}$	8	0
(ix)	$\ddot{x}+2\dot{x}+10x=e^{-t}\cos(3t)$	9	0
(x)	$\ddot{x}+6\dot{x}+10x=e^{-3t}\cos t$	10	0
(xi)	$\ddot{x}+4\dot{x}+4x=e^{2t}$	11	0

Strategy (Robinson Sec. 14.1, Lemma): the general solution of $L[x]=f$ is $x=x_h+x_p$, with x_h the general solution of $L[x]=0$ (two arbitrary constants c_1, c_2) and x_p any one particular solution. The constants are then fixed by the two initial conditions.

PHASE 2: METHOD, NOTATION AND THE GUESS FOR EACH ITEM

Write $P(k)=k^2+bk+c$ for the characteristic polynomial. A forcing $c_0\,t^m e^{\sigma t}\{\cos\omega t,\sin\omega t\}$ is, for the purpose of the guess, the exponent $s=\sigma+i\omega$ (and $s=\sigma$ when there is no sinusoid; $s=0$ for a polynomial). Rule of thumb (Robinson Sec. 14.5, p. 139): take the standard guess of the same form as f with undetermined coefficients, and multiply it by t^{m_s} where m_s is the multiplicity of s as a root of P ($m_s=0$ when $P(s)\neq 0$).

Forcing	Standard guess x_p	If s is a root of P of multiplicity m_s
polynomial of degree m	$K_0+K_1t+\cdots+K_mt^m$	multiply by t^{m_s}
$e^{\sigma t}$	$Ke^{\sigma t}$	$Kt^{m_s}e^{\sigma t}$
$\sin\omega t$ or $\cos\omega t$	$K_1\cos\omega t+K_2\sin\omega t$ (both, always)	$t^{m_s}(K_1\cos\omega t+K_2\sin\omega t)$
$e^{\sigma t}\cos\omega t$	$e^{\sigma t}(K_1\cos\omega t+K_2\sin\omega t)$	$t^{m_s}e^{\sigma t}(K_1\cos\omega t+K_2\sin\omega t)$

Applying the rule item by item (the roots of P were found in Exercise 4 of part 1 for the homogeneous equations; the last column is m_s):

Item	$P(k)$	Roots of P	Exponent s of f	m_s
(i)	k^2-4	$-2, 2$	0	0
(ii)	k^2-4k	$0, 4$	0	1
(iii)	k^2+k-2	$-2, 1$	-1	0
(iv)	k^2+k-2	$-2, 1$	1	1

Item	$P(k)$	Roots of P	Exponent s of f	m_s
(v)	k^2+2k+1	-1 (double)	-1	2
(vi)	$k^2+\omega^2$	$-i\omega, i\omega$	$i\alpha$	0
(vii)	$k^2+\omega^2$	$-i\omega, i\omega$	$i\omega$	1
(viii)	$k^2+2k+10$	$-1-3i, -1+3i$	-1	0
(ix)	$k^2+2k+10$	$-1-3i, -1+3i$	$-1+3i$	1
(x)	$k^2+6k+10$	$-3-i, -3+i$	$-3+i$	1
(xi)	k^2+4k+4	-2 (double)	2	0

Items with $m_s \geq 1$ (resonant forcing, marked by a non-zero last column) need the extra factor t^{m_s}: (ii), (iv), (v), (vii), (ix), (x).

PHASE 3: STEP-BY-STEP DERIVATION OF THE ELEVEN SOLUTIONS

(i) $\ddot{x} - 4x = t^2$, with $x(0) = 1$, $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} - 4x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 4 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(-4) = 16$.

Factorising, $P(k) = (k-2)\,(k+2)$, so the roots are

$$k = -2, \qquad k = 2$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, e^{2t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 e^{2t}$$

The forcing t^2 is a polynomial of degree 2; a polynomial is e^{0t} times a polynomial, so the exponent to test is $s = 0$: $P(0) = -4$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 + K_2 t + K_3 t^2$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_2 + 2K_3 t \\ x_p''(t) &= 2K_3 \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 4x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -4K_3\, t^2 - 4K_2\, t + (-4K_1 + 2K_3)$$

This must equal the forcing $f(t) = t^2$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} t^2: &\quad -4K_3 = 1 \\ t: &\quad -4K_2 = 0 \\ 1: &\quad -4K_1 + 2K_3 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} -4K_3 &= 1 \quad (1) \\ -4K_2 &= 0 \quad (2) \\ -4K_1 + 2K_3 &= 0 \quad (3) \end{aligned}$$

$-4K_3 = 1$ contains one unknown only, so $K_3 = -\frac{1}{4}$.

Substitute in $-4K_1 + 2K_3 = 0$: $-4K_1 - \frac{1}{2} = 0$.

$-4K_2 = 0$ contains one unknown only, so $K_2 = 0$.

$-4K_1 - \frac{1}{2} = 0$ contains one unknown only, so $K_1 = -\frac{1}{8}$.

Hence $K_1 = -\frac{1}{8}$, $K_2 = 0$, $K_3 = -\frac{1}{4}$.

Therefore

$$x_p(t) = -\frac{t^2}{4} - \frac{1}{8}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 e^{2t} - \frac{t^2}{4} - \frac{1}{8}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^{-2t} + 2c_2 e^{2t} - \frac{t}{2}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{1}{8} = 1$$
$$x'(0) = -2c_1 + 2c_2 = 0$$

The linear system is

$$c_1 + c_2 - \frac{1}{8} = 1 \quad (1)$$
$$-2c_1 + 2c_2 = 0 \quad (2)$$

Solve $c_1 + c_2 - \frac{1}{8} = 1$ for c_1: $c_1 = \frac{9}{8} - c_2$.

Substitute in $-2c_1 + 2c_2 = 0$: $4c_2 - \frac{9}{4} = 0$.

$4c_2 - \frac{9}{4} = 0$ contains one unknown only, so $c_2 = \frac{9}{16}$.

Back-substitution: $c_1 = \frac{9}{8} - c_2 = \frac{9}{16}$.

Hence $c_1 = \frac{9}{16}, c_2 = \frac{9}{16}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{t^2}{4} + \frac{9e^{2t}}{16} - \frac{1}{8} + \frac{9e^{-2t}}{16}}$$

(ii) $\ddot{x} - 4\dot{x} = t^2$**, with** $x(0) = 2,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} - 4\dot{x} = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 4k = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-4)^2 - 4(1)(0) = 16$.

Factorising, $P(k) = k\,(k-4)$, so the roots are

$$k = 0, \qquad k = 4$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $1,\, e^{4t}$, so the complementary function is

$$x_h(t) = c_1 + c_2 e^{4t}$$

The forcing t^2 is a polynomial of degree 2; a polynomial is e^{0t} times a polynomial, so the exponent to test is $s = 0$: $P(0) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t + K_2 t^2 + K_3 t^3$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= K_1 + 2K_2 t + 3K_3 t^2 \\ x_p''(t) &= 2K_2 + 6K_3 t \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 4x_p'$ and collect the terms that multiply the same function:

$$L[x_p] = -12K_3\,t^2 + (-8K_2 + 6K_3)\,t + (-4K_1 + 2K_2)$$

This must equal the forcing $f(t) = t^2$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} t^2: &\quad -12K_3 = 1 \\ t: &\quad -8K_2 + 6K_3 = 0 \\ 1: &\quad -4K_1 + 2K_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} -12K_3 &= 1 \quad (1) \\ -8K_2 + 6K_3 &= 0 \quad (2) \\ -4K_1 + 2K_2 &= 0 \quad (3) \end{aligned}$$

$-12K_3 = 1$ contains one unknown only, so $K_3 = -\frac{1}{12}$.

Substitute in $-8K_2 + 6K_3 = 0$: $-8K_2 - \frac{1}{2} = 0$.

$-8K_2 - \frac{1}{2} = 0$ contains one unknown only, so $K_2 = -\frac{1}{16}$.

Substitute in $-4K_1 + 2K_2 = 0$: $-4K_1 - \frac{1}{8} = 0$.

$-4K_1 - \frac{1}{8} = 0$ contains one unknown only, so $K_1 = -\frac{1}{32}$.

Hence $K_1 = -\frac{1}{32}, K_2 = -\frac{1}{16}, K_3 = -\frac{1}{12}$.

Therefore

$$x_p(t) = -\frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 + c_2 e^{4t} - \frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = 4c_2 e^{4t} - \frac{t^2}{4} - \frac{t}{8} - \frac{1}{32}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 = 2$$
$$x'(0) = 4c_2 - \frac{1}{32} = 0$$

The linear system is

$$c_1 + c_2 = 2 \quad (1)$$
$$4c_2 - \frac{1}{32} = 0 \quad (2)$$

$4c_2 - \frac{1}{32} = 0$ contains one unknown only, so $c_2 = \frac{1}{128}$.

Substitute in $c_1 + c_2 = 2$: $c_1 + \frac{1}{128} = 2$.

$c_1 + \frac{1}{128} = 2$ contains one unknown only, so $c_1 = \frac{255}{128}$.

Hence $c_1 = \frac{255}{128}, c_2 = \frac{1}{128}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32} + \frac{e^{4t}}{128} + \frac{255}{128}}$$

(iii) $\ddot{x} + \dot{x} - 2x = 3e^{-t}$**, with** $x(0) = 3,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \dot{x} - 2x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + k - 2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (1)^2 - 4(1)(-2) = 9$.

Factorising, $P(k) = (k - 1)\,(k + 2)$, so the roots are

$$k = -2, \qquad k = 1$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m - 1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, e^{t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 e^{t}$$

The forcing $3e^{-t}$ is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = -2$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -K_1 e^{-t}$$
$$x_p''(t) = K_1 e^{-t}$$

Substitute into $L[x_p] = x_p'' + x_p' - 2x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -2K_1\, e^{-t}$$

This must equal the forcing $f(t) = 3e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad -2K_1 = 3$$

The linear system is

$$-2K_1 = 3 \quad (1)$$

$-2K_1 = 3$ contains one unknown only, so $K_1 = -\frac{3}{2}$.

Hence $K_1 = -\frac{3}{2}$.

Therefore

$$x_p(t) = -\frac{3e^{-t}}{2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 e^{t} - \frac{3e^{-t}}{2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^{-2t} + c_2 e^{t} + \frac{3e^{-t}}{2}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{3}{2} = 3$$
$$x'(0) = -2c_1 + c_2 + \frac{3}{2} = 0$$

The linear system is

$$c_1 + c_2 - \frac{3}{2} = 3 \quad (1)$$
$$-2c_1 + c_2 + \frac{3}{2} = 0 \quad (2)$$

Solve $c_1 + c_2 - \frac{3}{2} = 3$ for c_1: $c_1 = \frac{9}{2} - c_2$.

Substitute in $-2c_1 + c_2 + \frac{3}{2} = 0$: $3c_2 - \frac{15}{2} = 0$.

$3c_2 - \frac{15}{2} = 0$ contains one unknown only, so $c_2 = \frac{5}{2}$.

Back-substitution: $c_1 = \frac{9}{2} - c_2 = 2$.

Hence $c_1 = 2, c_2 = \frac{5}{2}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{5e^{t}}{2} - \frac{3e^{-t}}{2} + 2e^{-2t}}$$

(iv) $\ddot{x} + \dot{x} - 2x = e^{t}$**, with** $x(0) = 4,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \dot{x} - 2x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^{2} + k - 2 = 0$$

Its discriminant is $\Delta = b^{2} - 4ac = (1)^{2} - 4(1)(-2) = 9$.

Factorising, $P(k) = (k - 1)\,(k + 2)$, so the roots are

$$k = -2, \qquad k = 1$$

A real root k of multiplicity m gives $t^{j}e^{kt}$ for $j = 0, \dots, m - 1$; a complex pair $\rho \pm i\omega$ gives $t^{j}e^{\rho t}\cos \omega t$ and $t^{j}e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, e^{t}, so the complementary function is

$$x_{h}(t) = c_{1}e^{-2t} + c_{2}e^{t}$$

The forcing e^{t} is an exponential with exponent $s = 1$; we test it in the characteristic polynomial: $P(1) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^{1}.

Guess for the particular integral, with undetermined coefficients:

$$x_{p}(t) = K_{1}te^{t}$$

Differentiate term by term (product rule on $t^{j}e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_{p}'(t) &= K_{1}te^{t} + K_{1}e^{t} \\ x_{p}''(t) &= K_{1}te^{t} + 2K_{1}e^{t} \end{aligned}$$

Substitute into $L[x_{p}] = x_{p}'' + x_{p}' - 2x_{p}$ and collect the terms that multiply the same function:

$$L[x_p] = 3K_1\, e^t$$

This must equal the forcing $f(t) = e^t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^t: \quad 3K_1 = 1$$

The linear system is

$$3K_1 = 1 \quad (1)$$

$3K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{3}$.

Hence $K_1 = \frac{1}{3}$.

Therefore

$$x_p(t) = \frac{te^t}{3}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1e^{-2t} + c_2e^t + \frac{te^t}{3}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1e^{-2t} + c_2e^t + \frac{te^t}{3} + \frac{e^t}{3}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 = 4$$
$$x'(0) = -2c_1 + c_2 + \frac{1}{3} = 0$$

The linear system is

$$c_1 + c_2 = 4 \quad (1)$$
$$-2c_1 + c_2 + \frac{1}{3} = 0 \quad (2)$$

Solve $c_1 + c_2 = 4$ for c_1: $c_1 = 4 - c_2$.

Substitute in $-2c_1 + c_2 + \frac{1}{3} = 0$: $3c_2 - \frac{23}{3} = 0$.

$3c_2 - \frac{23}{3} = 0$ contains one unknown only, so $c_2 = \frac{23}{9}$.

Back-substitution: $c_1 = 4 - c_2 = \frac{13}{9}$.

Hence $c_1 = \frac{13}{9}$, $c_2 = \frac{23}{9}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{t}}{3} + \frac{23e^{t}}{9} + \frac{13e^{-2t}}{9}}$$

(v) $\ddot{x} + 2\dot{x} + x = e^{-t}$**, with** $x(0) = 5,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 2\dot{x} + x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 2k + 1 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (2)^2 - 4(1)(1) = 0$.

Factorising, $P(k) = (k + 1)^2$, so the roots are

$$k = -1 \text{ (multiplicity } 2)$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m - 1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-t}, te^{-t}, so the complementary function is

$$x_h(t) = c_1 e^{-t} + c_2 te^{-t}$$

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 2, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^2.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t^2 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 t^2 e^{-t} + 2K_1 t e^{-t} \\ x_p''(t) &= K_1 t^2 e^{-t} - 4K_1 t e^{-t} + 2K_1 e^{-t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' + 2x_p' + x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_1\, e^{-t}$$

This must equal the forcing $f(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad 2K_1 = 1$$

The linear system is

$$2K_1 = 1 \quad (1)$$

$2K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{2}$.

Hence $K_1 = \frac{1}{2}$.

Therefore

$$x_p(t) = \frac{t^2 e^{-t}}{2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t} + c_2 t e^{-t} + \frac{t^2 e^{-t}}{2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1 e^{-t} - c_2 t e^{-t} + c_2 e^{-t} - \frac{t^2 e^{-t}}{2} + t e^{-t}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 = 5 \\ x'(0) &= -c_1 + c_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 &= 5 \quad (1) \\ -c_1 + c_2 &= 0 \quad (2) \end{aligned}$$

$c_1 = 5$ contains one unknown only, so $c_1 = 5$.

Substitute in $-c_1 + c_2 = 0$: $c_2 - 5 = 0$.

$c_2 - 5 = 0$ contains one unknown only, so $c_2 = 5$.

Hence $c_1 = 5, c_2 = 5$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{t^2 e^{-t}}{2} + 5 t e^{-t} + 5 e^{-t}}$$

(vi) $\ddot{x} + \omega^2 x = \sin \alpha t$**, with** $x(0) = 6$**,** $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \omega^2 x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + \omega^2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(\omega^2) = -4\omega^2$.

Factorising, $P(k) = (k - i\omega)\,(k + i\omega)$, so the roots are

$$k = -i\omega, \qquad k = i\omega$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $\cos\omega t$, $\sin\omega t$, so the complementary function is

$$x_h(t) = c_1\cos\omega t + c_2\sin\omega t$$

The forcing $\sin\alpha t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i\alpha)t}$, so we test $s = i\alpha$ in the characteristic polynomial: $P(i\alpha) = -\alpha^2 + \omega^2$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1\cos\alpha t + K_2\sin\alpha t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1\alpha\sin\alpha t + K_2\alpha\cos\alpha t \\ x_p''(t) &= -K_1\alpha^2\cos\alpha t - K_2\alpha^2\sin\alpha t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + \omega^2 x_p$ and collect the terms that multiply the same function:

$$L[x_p] = \left(-K_1\alpha^2 + K_1\omega^2\right)\cos\alpha\,t + \left(-K_2\alpha^2 + K_2\omega^2\right)\sin\alpha\,t$$

This must equal the forcing $f(t) = \sin\alpha t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos\alpha\,t: &\quad -K_1\alpha^2 + K_1\omega^2 = 0 \\ \sin\alpha\,t: &\quad -K_2\alpha^2 + K_2\omega^2 = 1 \end{aligned}$$

The linear system is

$$-K_1\alpha^2 + K_1\omega^2 = 0 \quad (1)$$
$$-K_2\alpha^2 + K_2\omega^2 = 1 \quad (2)$$

$-K_1\alpha^2 + K_1\omega^2 = 0$ contains one unknown only, so $K_1 = 0$.

$-K_2\alpha^2 + K_2\omega^2 = 1$ contains one unknown only, so $K_2 = -\frac{1}{\alpha^2-\omega^2}$.

Hence $K_1 = 0, K_2 = -\frac{1}{\alpha^2-\omega^2}$.

Therefore

$$x_p(t) = -\frac{\sin \alpha t}{\alpha^2 - \omega^2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 \cos \omega t + c_2 \sin \omega t - \frac{\sin \alpha t}{\alpha^2 - \omega^2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -\frac{\alpha \cos \alpha t}{\alpha^2 - \omega^2} - c_1\omega \sin \omega t + c_2\omega \cos \omega t$$

Evaluate at $t = 0$, using $e^0 = 1, \cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 6$$
$$x'(0) = -\frac{\alpha}{\alpha^2 - \omega^2} + c_2\omega = 0$$

The linear system is

$$c_1 = 6 \quad (1)$$
$$-\frac{\alpha}{\alpha^2 - \omega^2} + c_2\omega = 0 \quad (2)$$

$c_1 = 6$ contains one unknown only, so $c_1 = 6$.

$-\frac{\alpha}{\alpha^2-\omega^2} + c_2\omega = 0$ contains one unknown only, so $c_2 = \frac{\alpha}{\alpha^2\omega-\omega^3}$.

Hence $c_1 = 6, c_2 = \frac{\alpha}{\alpha^2\omega-\omega^3}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{\alpha \sin \omega t}{\alpha^2 \omega - \omega^3} + 6\cos \omega t - \frac{\sin \alpha t}{\alpha^2 - \omega^2}}$$

Writing the same solution with the denominator $\omega^2 - \alpha^2$ that appears naturally in the guess (check: $-\frac{1}{\alpha^2-\omega^2} = \frac{1}{\omega^2-\alpha^2}$ and $\frac{\alpha}{\alpha^2\omega-\omega^3} = -\frac{\alpha}{\omega(\omega^2-\alpha^2)}$):

$$\boxed{x(t) = 6\cos \omega t + \frac{\sin \alpha t - \dfrac{\alpha}{\omega}\sin \omega t}{\omega^2 - \alpha^2}}$$

The first term carries the initial displacement $x(0) = 6$; the fraction vanishes at $t = 0$ and has zero slope there ($\alpha \cos 0 - \alpha \cos 0 = 0$ after dividing by $\omega^2 - \alpha^2$), so the initial conditions hold.

(vii) $\ddot{x} + \omega^2 x = \sin \omega t$**, with** $x(0) = 7,\ \dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + \omega^2 x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + \omega^2 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (0)^2 - 4(1)(\omega^2) = -4\omega^2$.

Factorising, $P(k) = (k - i\omega)\,(k + i\omega)$, so the roots are

$$k = -i\omega, \qquad k = i\omega$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $\cos \omega t$, $\sin \omega t$, so the complementary function is

$$x_h(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

The forcing $\sin \omega t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i\omega)t}$, so we test $s = i\omega$ in the characteristic polynomial: $P(i\omega) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t \cos \omega t + K_2 t \sin \omega t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 \omega t \sin \omega t + K_1 \cos \omega t + K_2 \omega t \cos \omega t + K_2 \sin \omega t \\ x_p''(t) &= -K_1 \omega^2 t \cos \omega t - 2K_1 \omega \sin \omega t - K_2 \omega^2 t \sin \omega t + 2K_2 \omega \cos \omega t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + \omega^2 x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_2 \omega \, \cos \omega \, t - 2K_1 \omega \, \sin \omega \, t$$

This must equal the forcing $f(t) = \sin \omega t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos \omega \, t: &\quad 2K_2 \omega = 0 \\ \sin \omega \, t: &\quad -2K_1 \omega = 1 \end{aligned}$$

The linear system is

$$\begin{aligned} 2K_2 \omega &= 0 \quad (1) \\ -2K_1 \omega &= 1 \quad (2) \end{aligned}$$

$2K_2 \omega = 0$ contains one unknown only, so $K_2 = 0$.

$-2K_1 \omega = 1$ contains one unknown only, so $K_1 = -\frac{1}{2\omega}$.

Hence $K_1 = -\frac{1}{2\omega}$, $K_2 = 0$.

Therefore

$$x_p(t) = -\frac{t \cos \omega t}{2\omega}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 \cos \omega t + c_2 \sin \omega t - \frac{t \cos \omega t}{2\omega}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1 \omega \sin \omega t + c_2 \omega \cos \omega t + \frac{t \sin \omega t}{2} - \frac{\cos \omega t}{2\omega}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 = 7 \\ x'(0) &= c_2 \omega - \frac{1}{2\omega} = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 &= 7 \quad (1) \\ c_2 \omega - \frac{1}{2\omega} &= 0 \quad (2) \end{aligned}$$

$c_1 = 7$ contains one unknown only, so $c_1 = 7$.

$c_2 \omega - \frac{1}{2\omega} = 0$ contains one unknown only, so $c_2 = \frac{1}{2\omega^2}$.

Hence $c_1 = 7, c_2 = \frac{1}{2\omega^2}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = 7 \cos \omega t - \frac{t \cos \omega t}{2\omega} + \frac{\sin \omega t}{2\omega^2}}$$

Check of the resonant solution by the limit $\alpha \to \omega$ of the forced part of item (vi), the fraction, which does not depend on the initial data (L'Hopital's rule, differentiating numerator and denominator with respect to α):

$$\lim_{\alpha \to \omega} \frac{\sin \alpha t - \frac{\alpha}{\omega} \sin \omega t}{\omega^2 - \alpha^2} = \lim_{\alpha \to \omega} \frac{t \cos \alpha t - \frac{1}{\omega} \sin \omega t}{-2\alpha} = \frac{t \cos \omega t - \frac{1}{\omega} \sin \omega t}{-2\omega} = -\frac{t \cos \omega t}{2\omega} + \frac{\sin \omega t}{2\omega^2}$$

which is exactly the zero-data (forced) part of the resonant solution just found, i.e. everything except $7\cos\omega t$: the resonant solution is the limit of the non-resonant ones.

(viii) $\ddot{x}+2\dot{x}+10x=e^{-t}$**, with** $x(0)=8,\ \dot{x}(0)=0$

The homogeneous equation $\ddot{x}+2\dot{x}+10x=0$ has constant coefficients, so we try $x=e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt}\neq 0$ leaves the characteristic equation

$$P(k)=k^2+2k+10=0$$

Its discriminant is $\Delta=b^2-4ac=(2)^2-4(1)(10)=-36$.

Factorising, $P(k)=(k+1-3i)\,(k+1+3i)$, so the roots are

$$k=-1-3i,\qquad k=-1+3i$$

A real root k of multiplicity m gives t^je^{kt} for $j=0,\dots,m-1$; a complex pair $\rho\pm i\omega$ gives $t^je^{\rho t}\cos\omega t$ and $t^je^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-t}\cos 3t$, $e^{-t}\sin 3t$, so the complementary function is

$$x_h(t)=c_1e^{-t}\cos 3t+c_2e^{-t}\sin 3t$$

The forcing e^{-t} is an exponential with exponent $s=-1$; we test it in the characteristic polynomial: $P(-1)=9$.

Since $P(s)\neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t)=K_1e^{-t}$$

Differentiate term by term (product rule on $t^je^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$x_p'(t)=-K_1e^{-t}$$
$$x_p''(t)=K_1e^{-t}$$

Substitute into $L[x_p]=x_p''+2x_p'+10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 9K_1\, e^{-t}$$

This must equal the forcing $f(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad 9K_1 = 1$$

The linear system is

$$9K_1 = 1 \quad (1)$$

$9K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{9}$.

Hence $K_1 = \frac{1}{9}$.

Therefore

$$x_p(t) = \frac{e^{-t}}{9}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t + \frac{e^{-t}}{9}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^{-t}\sin 3t - c_1 e^{-t}\cos 3t - c_2 e^{-t}\sin 3t + 3c_2 e^{-t}\cos 3t - \frac{e^{-t}}{9}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + \frac{1}{9} = 8$$
$$x'(0) = -c_1 + 3c_2 - \frac{1}{9} = 0$$

The linear system is

$$c_1 + \frac{1}{9} = 8 \quad (1)$$
$$-c_1 + 3c_2 - \frac{1}{9} = 0 \quad (2)$$

$c_1 + \frac{1}{9} = 8$ contains one unknown only, so $c_1 = \frac{71}{9}$.

Substitute in $-c_1 + 3c_2 - \frac{1}{9} = 0$: $3c_2 - 8 = 0$.

$3c_2 - 8 = 0$ contains one unknown only, so $c_2 = \frac{8}{3}$.

Hence $c_1 = \frac{71}{9}$, $c_2 = \frac{8}{3}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{8e^{-t}\sin 3t}{3} + \frac{71e^{-t}\cos 3t}{9} + \frac{e^{-t}}{9}}$$

(ix) $\ddot{x} + 2\dot{x} + 10x = e^{-t}\cos 3t$**, with** $x(0) = 9$**,** $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 2\dot{x} + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 2k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (2)^2 - 4(1)(10) = -36$.

Factorising, $P(k) = (k + 1 - 3i)\,(k + 1 + 3i)$, so the roots are

$$k = -1 - 3i, \qquad k = -1 + 3i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-t}\cos 3t$, $e^{-t}\sin 3t$, so the complementary function is

$$x_h(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t$$

The forcing $e^{-t}\cos 3t$ is a sinusoidal term times an exponential: it is the real or imaginary part of $e^{(-1+3i)t}$, so we test $s = -1 + 3i$ in the characteristic polynomial: $P(-1 + 3i) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t e^{-t}\cos 3t + K_2 t e^{-t}\sin 3t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -3K_1 t e^{-t}\sin 3t - K_1 t e^{-t}\cos 3t + K_1 e^{-t}\cos 3t - K_2 t e^{-t}\sin 3t + 3K_2 t e^{-t}\cos 3t + K_2 e^{-t}\sin 3t$$
$$x_p''(t) = 6K_1 t e^{-t}\sin 3t - 8K_1 t e^{-t}\cos 3t - 6K_1 e^{-t}\sin 3t - 2K_1 e^{-t}\cos 3t - 8K_2 t e^{-t}\sin 3t - 6K_2 t e^{-t}\cos 3t - 2K_2 e^{-t}\sin 3t + 6K_2 e^{-t}\cos 3t$$

Substitute into $L[x_p] = x_p'' + 2x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 6K_2\, e^{-t}\cos 3t - 6K_1\, e^{-t}\sin 3t$$

This must equal the forcing $f(t) = e^{-t}\cos 3t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} e^{-t}\cos 3t: &\quad 6K_2 = 1 \\ e^{-t}\sin 3t: &\quad -6K_1 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} 6K_2 &= 1 \quad (1) \\ -6K_1 &= 0 \quad (2) \end{aligned}$$

$6K_2 = 1$ contains one unknown only, so $K_2 = \frac{1}{6}$.

$-6K_1 = 0$ contains one unknown only, so $K_1 = 0$.

Hence $K_1 = 0$, $K_2 = \frac{1}{6}$.

Therefore

$$x_p(t) = \frac{t e^{-t}\sin 3t}{6}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t}\cos 3t + c_2 e^{-t}\sin 3t + \frac{te^{-t}\sin 3t}{6}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^{-t}\sin 3t - c_1 e^{-t}\cos 3t - c_2 e^{-t}\sin 3t + 3c_2 e^{-t}\cos 3t - \frac{te^{-t}\sin 3t}{6} + \frac{te^{-t}\cos 3t}{2} + \frac{e^{-t}\sin 3t}{6}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 = 9 \\ x'(0) &= -c_1 + 3c_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 &= 9 \quad (1) \\ -c_1 + 3c_2 &= 0 \quad (2) \end{aligned}$$

$c_1 = 9$ contains one unknown only, so $c_1 = 9$.

Substitute in $-c_1 + 3c_2 = 0$: $3c_2 - 9 = 0$.

$3c_2 - 9 = 0$ contains one unknown only, so $c_2 = 3$.

Hence $c_1 = 9, c_2 = 3$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{-t}\sin 3t}{6} + 3e^{-t}\sin 3t + 9e^{-t}\cos 3t}$$

(x) $\ddot{x} + 6\dot{x} + 10x = e^{-3t}\cos t$**, with** $x(0) = 10$**,** $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 6\dot{x} + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 6k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (6)^2 - 4(1)(10) = -4$.

Factorising, $P(k) = (k + 3 - i)\,(k + 3 + i)$, so the roots are

$$k = -3 - i, \qquad k = -3 + i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-3t}\cos t$, $e^{-3t}\sin t$, so the complementary function is

$$x_h(t) = c_1 e^{-3t}\cos t + c_2 e^{-3t}\sin t$$

The forcing $e^{-3t}\cos t$ is a sinusoidal term times an exponential: it is the real or imaginary part of $e^{(-3+i)t}$, so we test $s = -3 + i$ in the characteristic polynomial: $P(-3+i) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t e^{-3t}\cos t + K_2 t e^{-3t}\sin t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 t e^{-3t}\sin t - 3K_1 t e^{-3t}\cos t + K_1 e^{-3t}\cos t - 3K_2 t e^{-3t}\sin t + K_2 t e^{-3t}\cos t + K_2 e^{-3t}\sin t \\ x_p''(t) &= 6K_1 t e^{-3t}\sin t + 8K_1 t e^{-3t}\cos t - 2K_1 e^{-3t}\sin t - 6K_1 e^{-3t}\cos t + 8K_2 t e^{-3t}\sin t - 6K_2 t e^{-3t}\cos t - 6K_2 e^{-3t}\sin t + 2K_2 e^{-3t}\cos t \end{aligned}$$

Substitute into $L[x_p] = x_p'' + 6x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_2\, e^{-3t}\cos t - 2K_1\, e^{-3t}\sin t$$

This must equal the forcing $f(t) = e^{-3t}\cos t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} e^{-3t}\cos t: &\quad 2K_2 = 1 \\ e^{-3t}\sin t: &\quad -2K_1 = 0 \end{aligned}$$

The linear system is

$$2K_2 = 1 \quad (1)$$
$$-2K_1 = 0 \quad (2)$$

$2K_2 = 1$ contains one unknown only, so $K_2 = \frac{1}{2}$.

$-2K_1 = 0$ contains one unknown only, so $K_1 = 0$.

Hence $K_1 = 0, K_2 = \frac{1}{2}$.

Therefore

$$x_p(t) = \frac{te^{-3t}\sin t}{2}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1e^{-3t}\cos t + c_2e^{-3t}\sin t + \frac{te^{-3t}\sin t}{2}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1e^{-3t}\sin t - 3c_1e^{-3t}\cos t - 3c_2e^{-3t}\sin t + c_2e^{-3t}\cos t - \frac{3te^{-3t}\sin t}{2} + \frac{te^{-3t}\cos t}{2} + \frac{e^{-3t}\sin t}{2}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 10$$
$$x'(0) = -3c_1 + c_2 = 0$$

The linear system is

$$c_1 = 10 \quad (1)$$
$$-3c_1 + c_2 = 0 \quad (2)$$

$c_1 = 10$ contains one unknown only, so $c_1 = 10$.

Substitute in $-3c_1 + c_2 = 0$: $c_2 - 30 = 0$.

$c_2 - 30 = 0$ contains one unknown only, so $c_2 = 30$.

Hence $c_1 = 10, c_2 = 30$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{-3t}\sin t}{2} + 30e^{-3t}\sin t + 10e^{-3t}\cos t}$$

(xi) $\ddot{x} + 4\dot{x} + 4x = e^{2t}$**, with** $x(0) = 11$**,** $\dot{x}(0) = 0$

The homogeneous equation $\ddot{x} + 4\dot{x} + 4x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + 4k + 4 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (4)^2 - 4(1)(4) = 0$.

Factorising, $P(k) = (k + 2)^2$, so the roots are

$$k = -2 \text{ (multiplicity } 2)$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2t}, te^{-2t}, so the complementary function is

$$x_h(t) = c_1 e^{-2t} + c_2 te^{-2t}$$

The forcing e^{2t} is an exponential with exponent $s = 2$; we test it in the characteristic polynomial: $P(2) = 16$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{2t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = 2K_1 e^{2t}$$
$$x_p''(t) = 4K_1 e^{2t}$$

Substitute into $L[x_p] = x_p'' + 4x_p' + 4x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 16K_1\, e^{2t}$$

This must equal the forcing $f(t) = e^{2t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{2t}: \quad 16K_1 = 1$$

The linear system is

$$16K_1 = 1 \quad (1)$$

$16K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{16}$.

Hence $K_1 = \frac{1}{16}$.

Therefore

$$x_p(t) = \frac{e^{2t}}{16}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-2t} + c_2 t e^{-2t} + \frac{e^{2t}}{16}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^{-2t} - 2c_2 t e^{-2t} + c_2 e^{-2t} + \frac{e^{2t}}{8}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + \frac{1}{16} = 11$$
$$x'(0) = -2c_1 + c_2 + \frac{1}{8} = 0$$

The linear system is

$$c_1 + \frac{1}{16} = 11 \quad (1)$$
$$-2c_1 + c_2 + \frac{1}{8} = 0 \quad (2)$$

$c_1 + \frac{1}{16} = 11$ contains one unknown only, so $c_1 = \frac{175}{16}$.

Substitute in $-2c_1 + c_2 + \frac{1}{8} = 0$: $c_2 - \frac{87}{4} = 0$.

$c_2 - \frac{87}{4} = 0$ contains one unknown only, so $c_2 = \frac{87}{4}$.

Hence $c_1 = \frac{175}{16}, c_2 = \frac{87}{4}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{87te^{-2t}}{4} + \frac{e^{2t}}{16} + \frac{175e^{-2t}}{16}}$$

PHASE 4: INTERPRETATION, LONG-TIME BEHAVIOUR AND CHECKS

Summary of the eleven solutions of the IVPs (item (n) has $x(0) = n, \dot{x}(0) = 0$):

Item	**Solution** $x(t)$
(i)	$-\frac{t^2}{4} + \frac{9e^{2t}}{16} - \frac{1}{8} + \frac{9e^{-2t}}{16}$
(ii)	$-\frac{t^3}{12} - \frac{t^2}{16} - \frac{t}{32} + \frac{e^{4t}}{128} + \frac{255}{128}$
(iii)	$\frac{5e^{t}}{2} - \frac{3e^{-t}}{2} + 2e^{-2t}$
(iv)	$\frac{te^{t}}{3} + \frac{23e^{t}}{9} + \frac{13e^{-2t}}{9}$
(v)	$\frac{t^2e^{-t}}{2} + 5te^{-t} + 5e^{-t}$
(vi)	$6\cos\omega t + \dfrac{\sin\alpha t - (\alpha/\omega)\sin\omega t}{\omega^2 - \alpha^2}$
(vii)	$7\cos\omega t - \frac{t\cos\omega t}{2\omega} + \frac{\sin\omega t}{2\omega^2}$
(viii)	$\frac{8e^{-t}\sin 3t}{3} + \frac{71e^{-t}\cos 3t}{9} + \frac{e^{-t}}{9}$
(ix)	$\frac{te^{-t}\sin 3t}{6} + 3e^{-t}\sin 3t + 9e^{-t}\cos 3t$
(x)	$\frac{te^{-3t}\sin t}{2} + 30e^{-3t}\sin t + 10e^{-3t}\cos t$

Item	Solution $x(t)$
(xi)	$\frac{87te^{-2t}}{4} + \frac{e^{2t}}{16} + \frac{175e^{-2t}}{16}$

Behaviour as $t \to \infty$

- (i): all terms of the complementary function except the growing one are small: $x \sim \frac{9}{16}e^{2t}$, unbounded.
- (ii): the term $\frac{1}{128}e^{4t}$ is tiny at first but eventually dominates the cubic polynomial: unbounded.
- (iii): $x \sim \frac{5}{2}e^{t}$ (the forcing $3e^{-t}$ and the mode e^{-2t} decay): unbounded.
- (iv): resonance with the root $k = 1$: $x \sim (\frac{t}{3} + \frac{23}{9})e^{t}$, faster than a pure exponential.
- (v): every term carries e^{-t}: $x \to 0$, although the resonant term $\frac{1}{2}t^2e^{-t}$ first grows (critical damping, resonant forcing).
- (vi): bounded for all t (a sum of two sinusoids, periodic exactly when α/ω is rational); no decay; beat-like modulation appears when $\alpha \approx \omega$.
- (vii): resonance: the amplitude of $-\dfrac{t\cos\omega t}{2\omega}$ grows linearly without bound.
- (viii): all terms decay like e^{-t}: $x \to 0$; the non-oscillatory forced part $\frac{1}{9}e^{-t}$ is not resonant.
- (ix): resonance with the damped oscillation: the envelope $\frac{t}{6}e^{-t}$ peaks at $t = 1$ (value $\frac{1}{6e} \approx 0.061$) and then decays; $x \to 0$.
- (x): resonance with $e^{-3t}\cos t$: the envelope $\frac{t}{2}e^{-3t}$ peaks at $t = \frac{1}{3}$ and decays; $x \to 0$.
- (xi): $x \sim \frac{1}{16}e^{2t}$: the forced exponential beats the decaying modes e^{-2t}, te^{-2t}: unbounded.

Checks

- Every solution was substituted back into its equation and into both initial conditions symbolically (script `scripts/verify_topic3_topic4.py`): the residual $L[x] - f$ is identically zero and $x(0) = n$, $\dot{x}(0) = 0$.
- Dimensions: if x is a length and t a time, every term of $\ddot{x} + b\dot{x} + cx = f$ has units of length/time^2, so b has units s^{-1}, c and ω^2 units s^{-2} and f units m/s^2 if the equation is read dimensionally. The sheet itself uses dimensionless variables (forcings such as t^2 or e^{-t} carry no units), so these units only serve to check the structure of the formulas. In (vi) the combination α/ω is dimensionless and $1/(\omega^2 - \alpha^2)$ has units s^2, so x comes out in metres.
- Resonance diagnostic: the items with a factor t in the particular integral are exactly those where the exponent of f is a root of P (listed in Phase 2).
- Beats versus resonance: as $\alpha \to \omega$, (vi) tends to (vii) (L'Hopital check above); the beat amplitude $\propto 1/(\omega^2 - \alpha^2)$ blows up exactly where the resonant solution replaces it.